

Ramaiah Institute of Technology

Department of AI & DS

Data Communication & Networking

Unit 2

Chapter 3

Introduction to Physical Layer

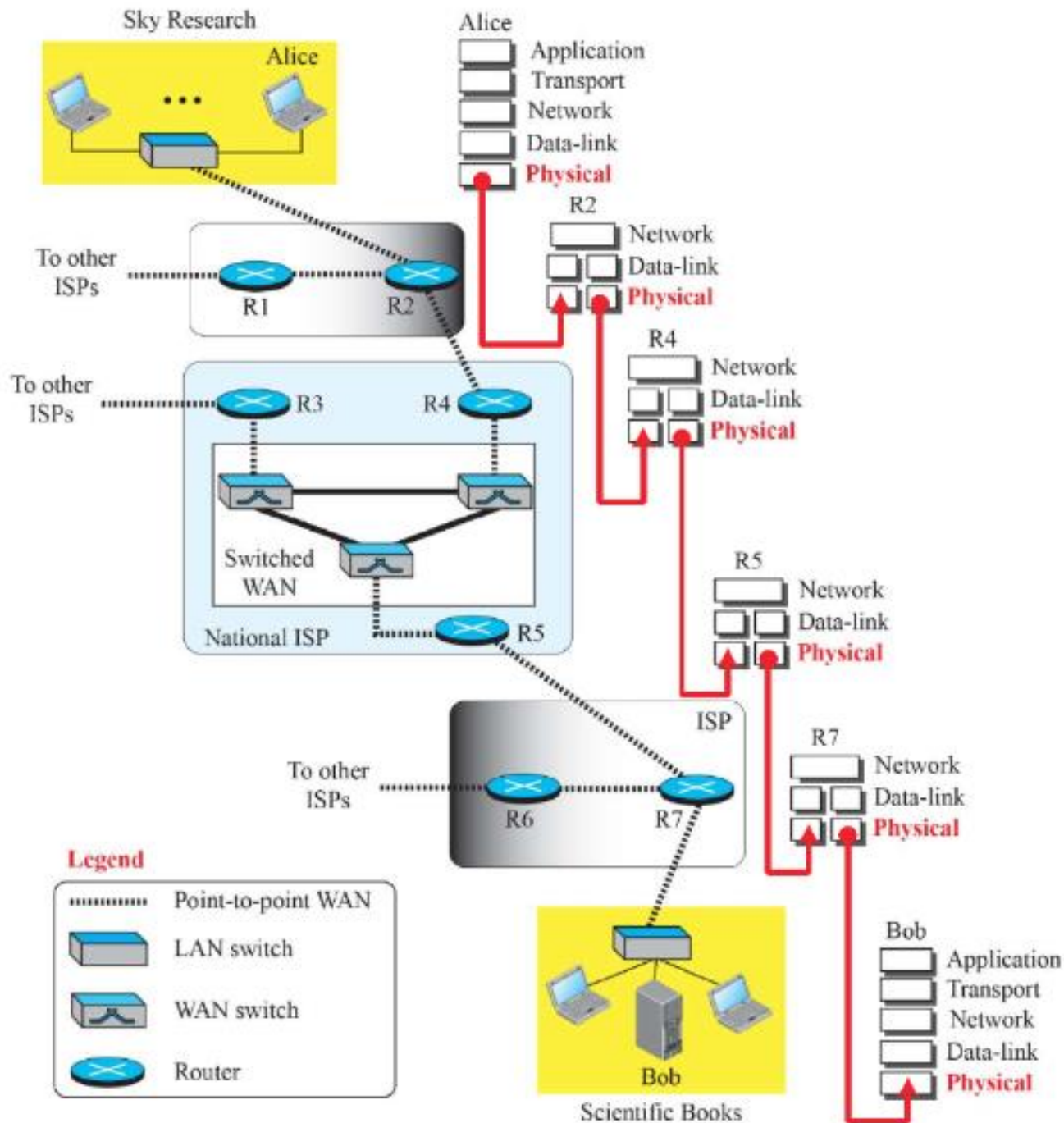
Data And Signals

- *One of the major functions of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.*
- *Whether you are collecting numerical statistics from another computer, sending animated pictures from a design workstation, or causing a bell to ring at a distant control center, you are working with the transmission of data across network connections.*
- *Transmission media work by conducting energy along a physical path. For transmission, data needs to be changed to signals.*

Data And Signals

Figure 3.1 shows a scenario in which a scientist working in a research company, Sky Research, needs to order a book related to her research from an online bookseller, Scientific Books.

Data



Analog and Digital Data

- *Analog data : Information that is continuous*
 - *Example: an analog clock that has hour, minute, and second hands gives information in a continuous form. The movements of the hands are continuous.*
- *Digital data: Information that has discrete states.*
 - *Example: a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.*

Analog and Digital Signals

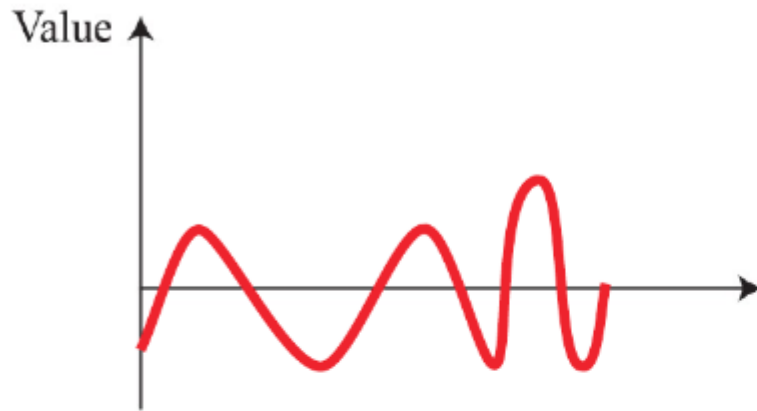
Like the data they represent, signals can be either analog or digital.

An analog signal has infinitely many levels of intensity over a period of time. As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path.

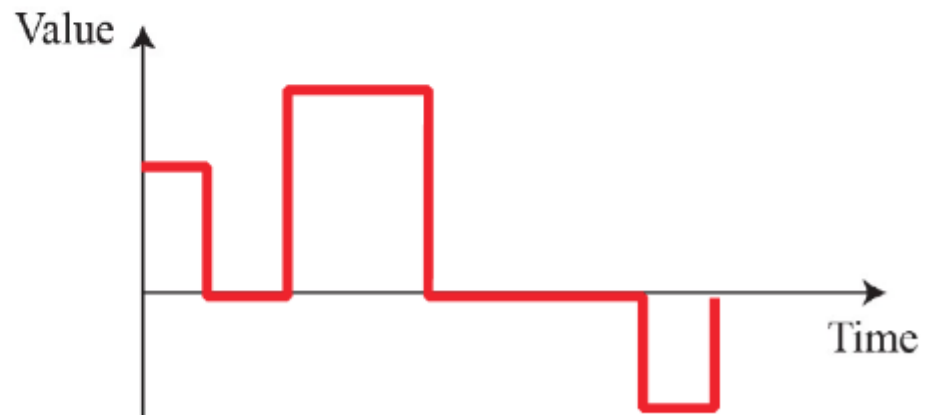
A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.

Analog and Digital Signals

Figure 3.2: Comparison of analog and digital signals



a. Analog signal



b. Digital signal

Periodic and Non Periodic Signals

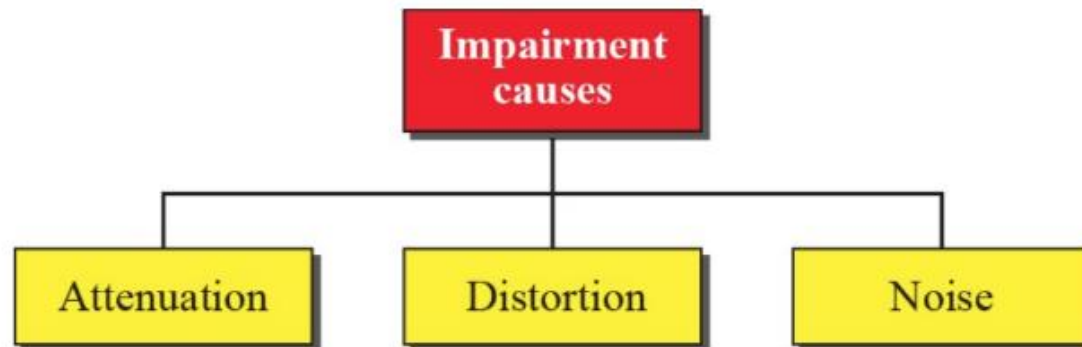
A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a cycle.

A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.

Transmission Impairment

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- Signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.
- Three causes of impairment are

Figure 3.26: Causes of impairment

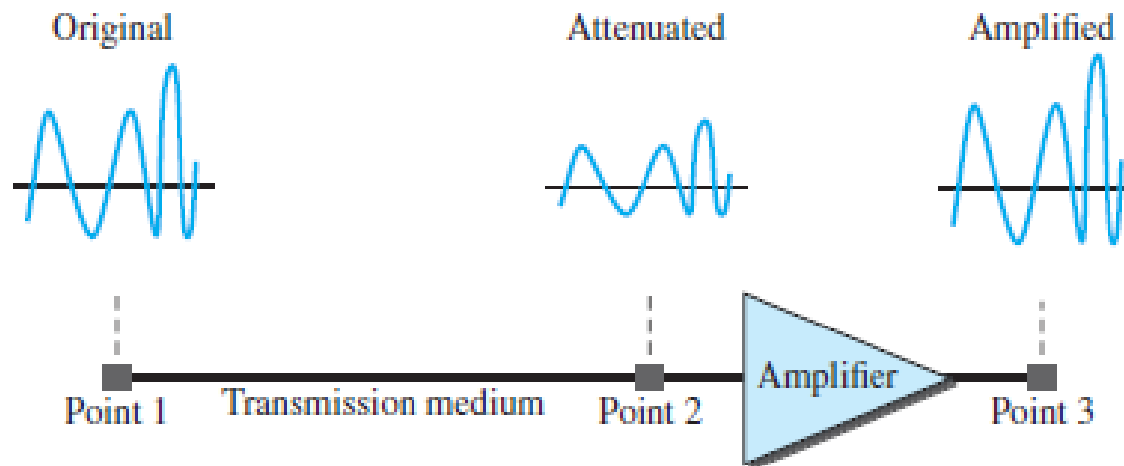


Transmission Impairment-Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- A wire carrying electric signals gets warm, if not hot, after a while.
- Some of the **electrical energy in the signal is converted to heat.**
- To compensate for this loss, amplifiers are used to amplify the signal.

Transmission Impairment-Attenuation

Figure 3.27 Attenuation



Transmission Impairment-Attenuation

- To show that a signal **has lost or gained strength**, engineers use the unit of the **decibel**.
- The decibel (dB) measures the **relative strengths** of two signals or one signal at two different points.
- **The decibel is negative if a signal is attenuated and positive if a signal is amplified.**

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

- Variables P1 and P2 are the powers of a signal at points 1 and 2.
- If the decibel is in terms of voltage instead of power, then power is proportional to the square of the voltage.

$$\text{dB} = 20 \log_{10} (V_2/V_1).$$

Transmission Impairment-Attenuation

Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that $P_2 = \frac{1}{2} P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Example 3.27

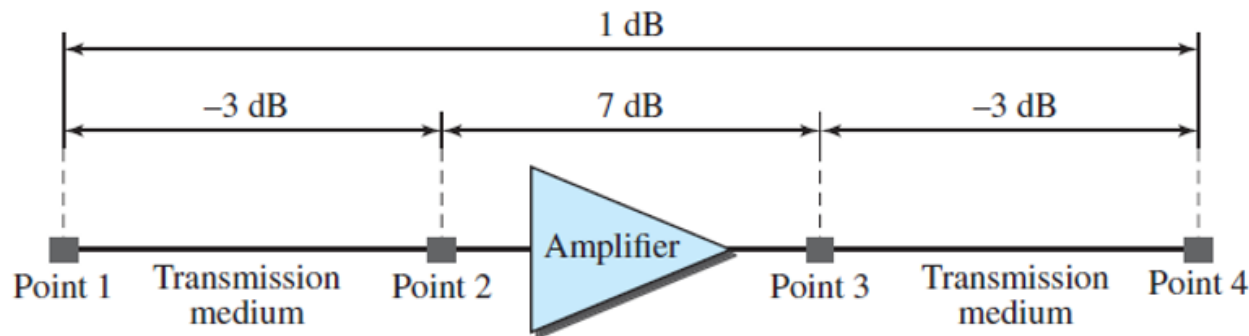
A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Transmission Impairment-Attenuation

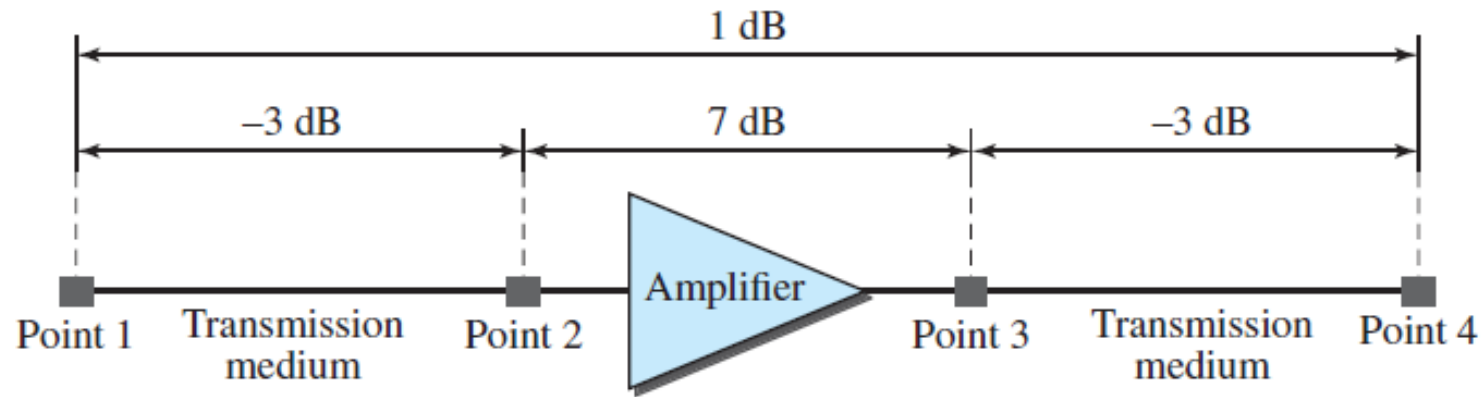
- The decibel measures can be added (or subtracted) when we are measuring several points (cascading) instead of just two.
- Example: 3.28
 - A signal travels from point 1 to point 4. The signal is attenuated by the time it reaches point 2. Between points 2 and 3, the signal is amplified.
 - Between points 3 and 4, the signal is attenuated.
 - Resultant decibel value for the signal just by adding the decibel measurements between each set of points.

Figure 3.28 *Decibels for Example 3.28*



Transmission Impairment-Attenuation

Figure 3.28 *Decibels for Example 3.28*



In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

The signal has gained in power

Transmission Impairment-Attenuation

Example 3.30

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} (P_2 / P_1) = -1.5 \quad \longrightarrow \quad (P_2 / P_1) = 10^{-0.15} = 0.71$$

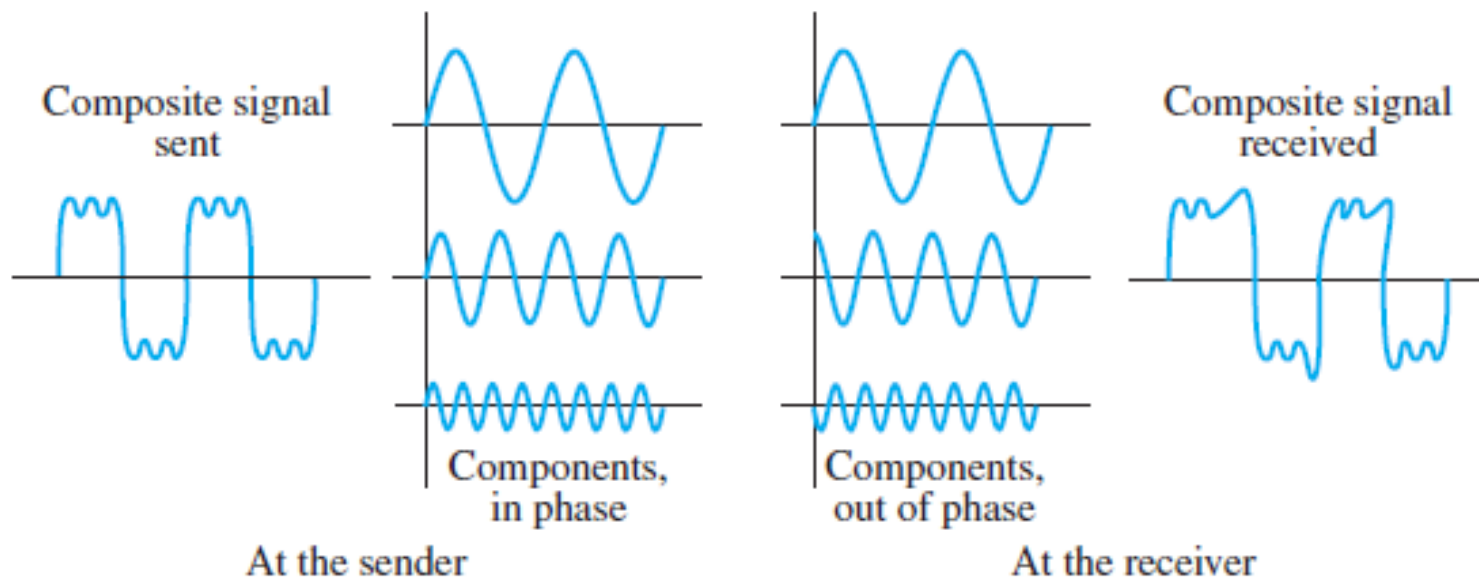
$$P_2 = 0.71P_1 = 0.7 \times 2 \text{ mW} = 1.4 \text{ mW}$$

Transmission Impairment-Distortion

- **Distortion** means that the **signal changes its form or shape**.
- Distortion can occur in a composite signal made of different frequencies.
- Each signal component has its own **propagation speed** through a medium and, its **own delay** in arriving at the final destination.
- Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.
- Signal components at the receiver have phases different from what they had at the sender.

Transmission Impairment-Distortion

Figure 3.29 *Distortion*

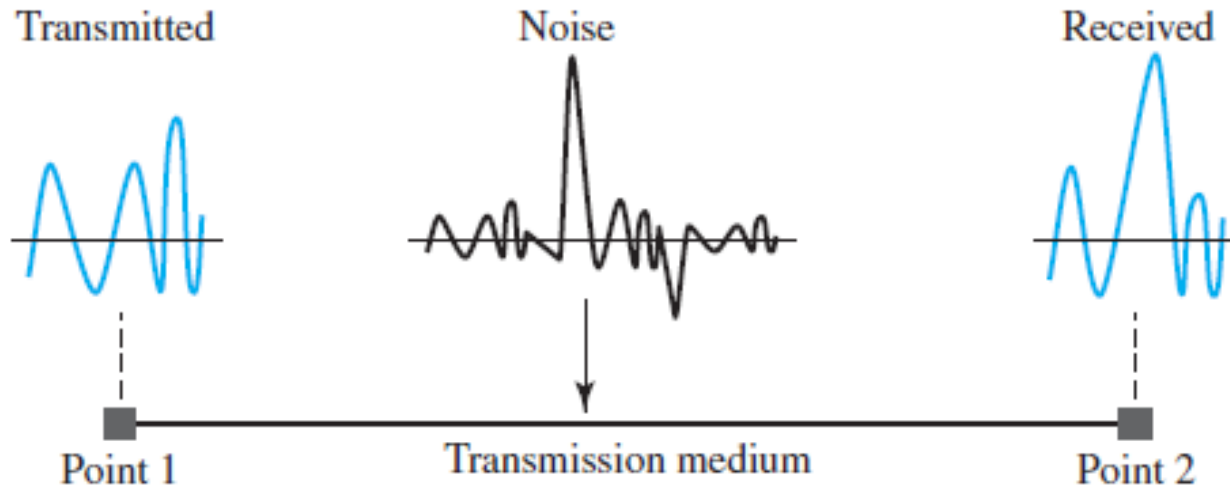


Transmission Impairment-Noise

- Noise is another cause of impairment.
- **Types of noise**, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- **Thermal noise** is the **random motion of electrons** in a wire, which creates an **extra signal** not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances.
- These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna

Transmission Impairment-Noise

Figure 3.30 *Noise*



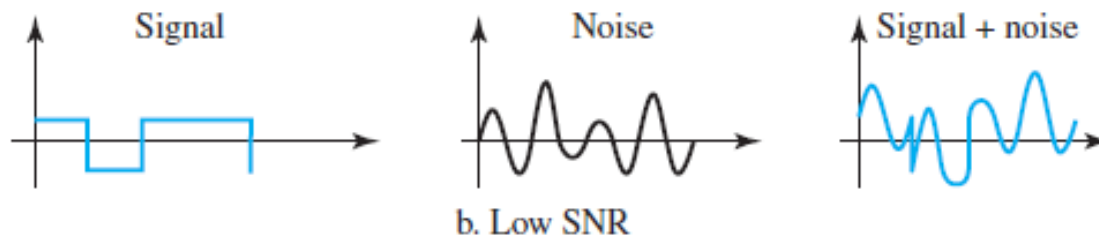
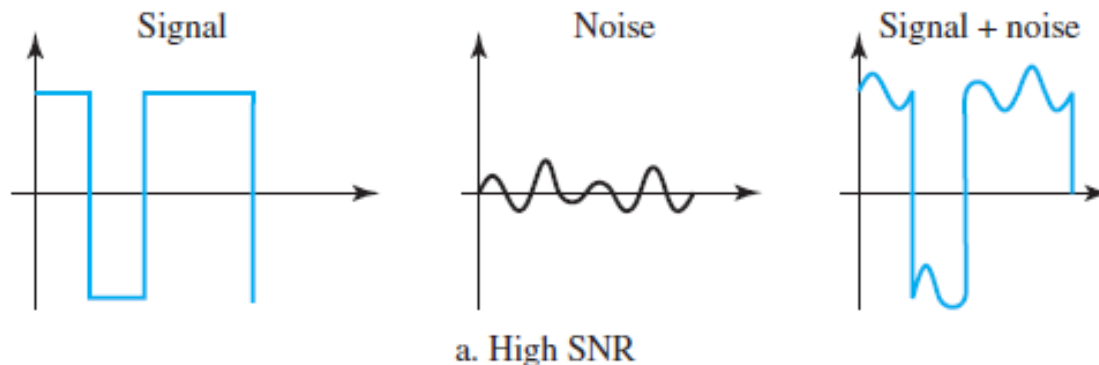
Transmission Impairment-Noise

Signal-to-Noise Ratio (SNR)

Signal-to-noise ratio is defined as

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

Figure 3.31 *Two cases of SNR: a high SNR and a low SNR*



Transmission Impairment-Noise

- SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
- A **high SNR** means the signal is less corrupted by noise
- A **low SNR** means the signal is more corrupted by noise.
- Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB} , defined as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

Transmission Impairment-Noise

Example 3.31

The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SNR_{dB}?

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$\text{SNR} = (10,000 \mu\text{w}) / (1 \mu\text{w}) = 10,000 \quad \text{SNR}_{\text{dB}} = 10 \log_{10} 10,000 = 10 \log_{10} 10^4 = 40$$

Example 3.32

The values of SNR and SNR_{dB} for a noiseless channel are

$$\text{SNR} = (\text{signal power}) / 0 = \infty \longrightarrow \text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

Data Rate Limits

- A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:
 1. The bandwidth available
 2. The level of the signals
 3. The quality of the channel (the level of noise)
- Two theoretical formulas were developed to calculate the data rate:
 1. By Nyquist for a noiseless channel,
 2. By Shannon for a noisy channel.

Data Rate Limits : Noiseless Channel: Nyquist Bit Rate

- For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- Bandwidth is the bandwidth of the channel,
- L is the number of signal levels used to represent data
- BitRate is the bit rate in bits per second.

Data Rate Limits : Noiseless Channel: Nyquist Bit Rate

- Given a specific bandwidth, Increase the number of signal levels, impose a burden on the receiver.
- **Ex:** If the number of levels in a signal is just **2**, the receiver can easily distinguish between a 0 and a 1.
- If the level of a signal is **64**, the receiver must be very sophisticated to distinguish between 64 different levels.
- In other words, increasing the levels of a signal reduces the reliability of the system.
- **Increasing the levels of a signal may reduce the reliability of the system.**

Data Rate Limits : Noiseless Channel: Nyquist Bit Rate

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12,000 \text{ bps}$$

Data Rate Limits : Noiseless Channel: Nyquist Bit Rate

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265,000 = 2 \times 20,000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Data Rate Limits : Noisy Channel: Shannon Capacity

- In reality, we cannot have a noiseless channel; the channel is always noisy.
- In 1944, Claude Shannon introduced a formula, called the **Shannon capacity**, to determine the **theoretical highest data rate for a noisy channel**:

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Data Rate Limits : Noisy Channel: Shannon Capacity

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

- **Bandwidth** is the bandwidth of the channel,
 - **SNR** is the signal-to-noise ratio,
 - **Capacity** is the capacity of the channel in bits per second.
-
- In Shannon formula there is no indication of the signal level, which means that no matter how many levels.
 - Cannot achieve a data rate higher than the capacity of the channel.

Data Rate Limits : Noisy Channel: Shannon Capacity

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Data Rate Limits : Noisy Channel: Shannon Capacity

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34,860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Data Rate Limits : Using Both Limits

Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2(1 + \text{SNR}) = 10^6 \log_2(1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \longrightarrow L = 4$$

The Shannon capacity gives us the upper limit;
the Nyquist formula tells us how many signal levels we need.

Data Rate Limits : Formulas

Noiseless: Nyquist Bit Rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

Noisy Channel: Shannon Capacity

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Additional Problems

1. A signal travels from point A to point B. At point A, the signal power is 100 W. At point B, the power is 90 W. What is the attenuation in decibels?

$$\text{dB} = 10 \log_{10} (90 / 100) = -0.46 \text{ dB}$$

2. The attenuation of a signal is -10 dB. What is the final signal power if it was originally 5 W?

$$-10 = 10 \log_{10} (P_2 / 5) \rightarrow \log_{10} (P_2 / 5) = -1 \rightarrow (P_2 / 5) = 10^{-1} \rightarrow P_2 = 0.5 \text{ W}$$

3. If the bandwidth of the channel is 5 Kbps, how long does it take to send a frame of 100,000 bits out of this device?

$$100,000 \text{ bits} / 5 \text{ Kbps} = 20 \text{ s}$$

Additional Problems

4. A line has a signal-to-noise ratio of 1000 and a bandwidth of 4000 KHz. What is the maximum data rate supported by this line?

$$4,000 \log_2 (1 + 1,000) \approx 40 \text{ Kbps}$$

5. We measure the performance of a telephone line (4 KHz of bandwidth). When the signal is 10 V, the noise is 5 mV. What is the maximum data rate supported by this telephone line?

$$4,000 \log_2 (1 + 10 / 0.005) = 43,866 \text{ bps}$$

6. A file contains 2 million bytes. How long does it take to download this file using a 56-Kbps channel? 1-Mbps channel?

The file contains $2,000,000 \times 8 = 16,000,000$ bits. With a 56-Kbps channel, it takes $16,000,000 / 56,000 = 289 \text{ s}$. With a 1-Mbps channel, it takes 16 s .

Chapter 4

Digital to Digital Conversion

Digital-to-Digital Conversion

How we can represent digital data by using digital signals?

The conversion involves three techniques:

- *line coding*
- *block coding*
- *scrambling*

Line coding is always needed; block coding and scrambling may or may not be needed.

Line Coding

Line coding is the process of converting digital data to digital signals.

We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits

- *Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal. Figure 4.1 shows the process.*

Line Coding

In telecommunication, a line code is a **pattern** of voltage, current, or photons used to represent digital data transmitted down a communication channel.

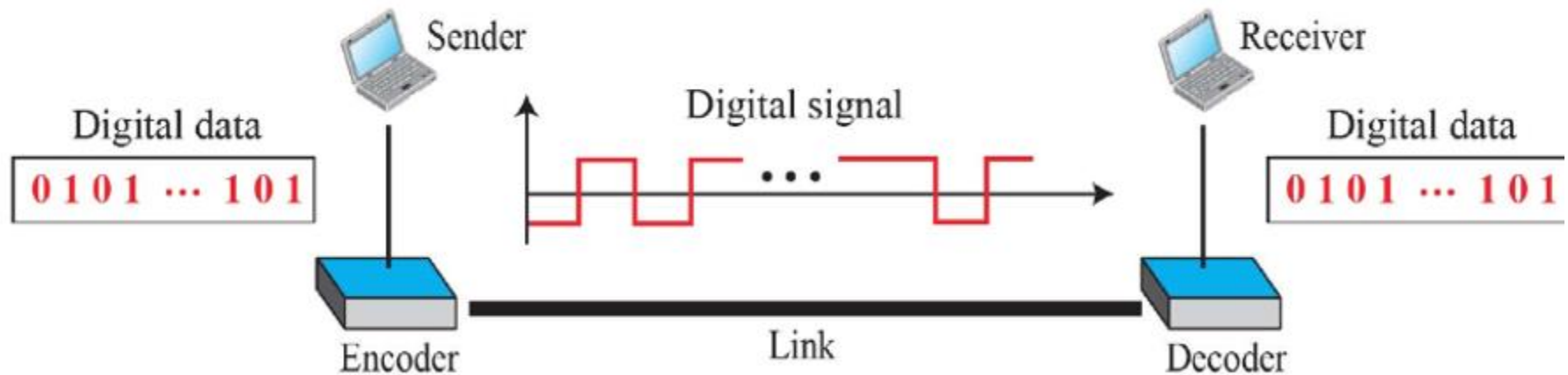
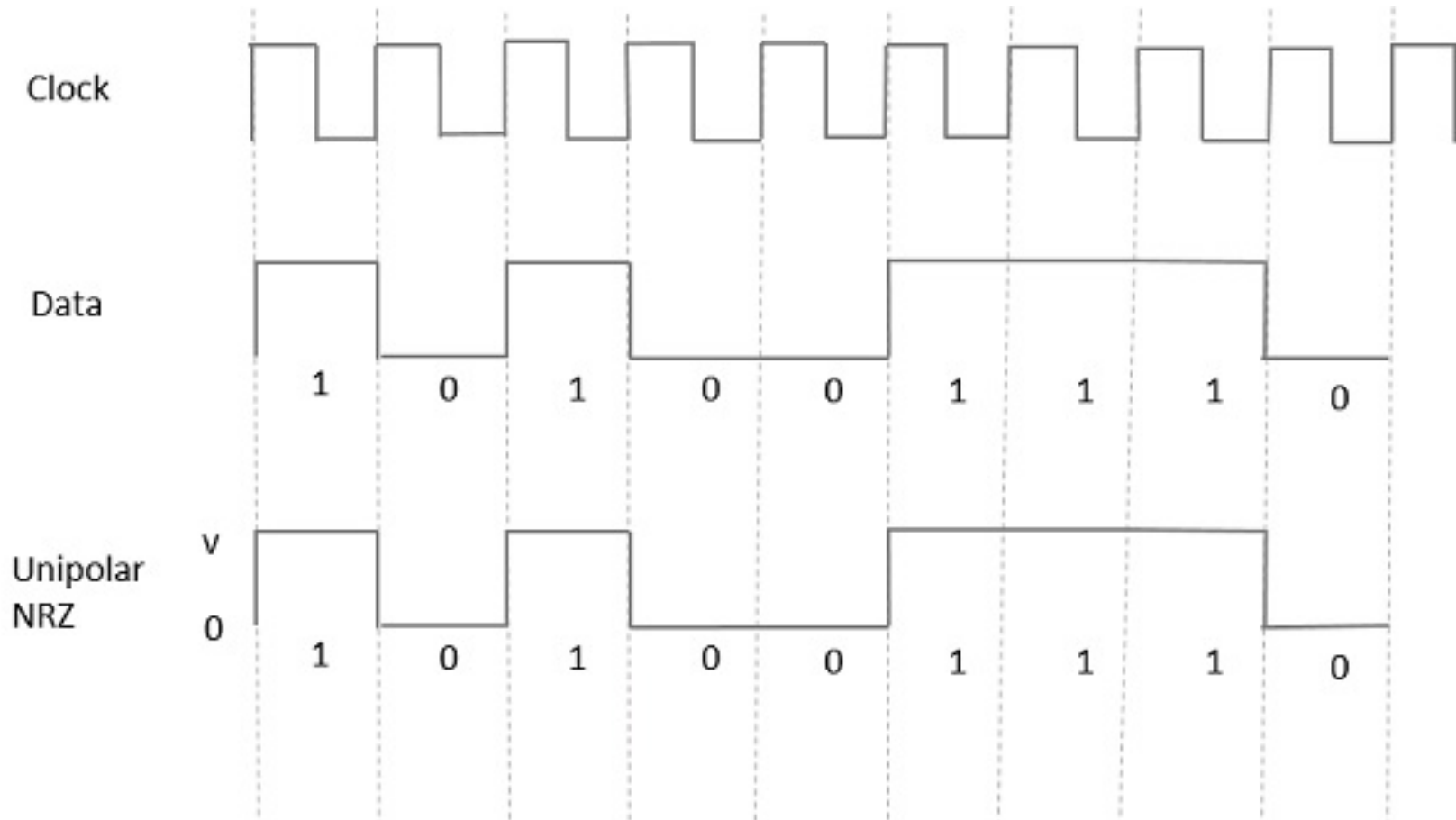


Figure 4.1: Line coding and decoding

Line Coding



Line Coding

Signal Element Versus Data Element

- In data communications, our goal is to send data elements. A data element is the smallest entity that can represent a piece of information: this is the bit.
- In digital data communications, a **signal element carries data elements.**
- A signal element is the shortest unit (timewise) of a digital signal.
- In other words, data elements are what we need to send; signal elements are what we can send.
- **Data elements are being carried; signal elements are the carriers.**

Line Coding

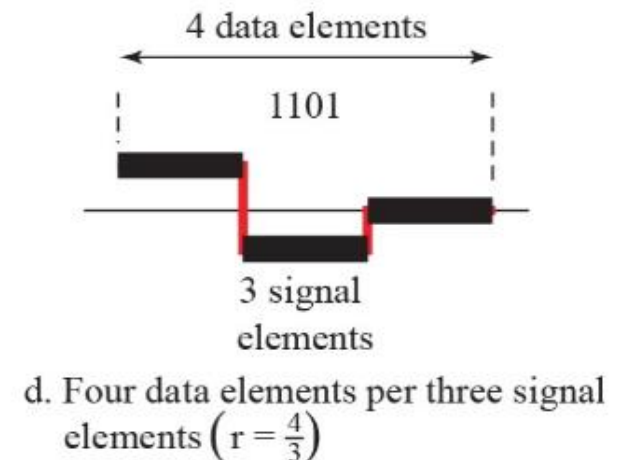
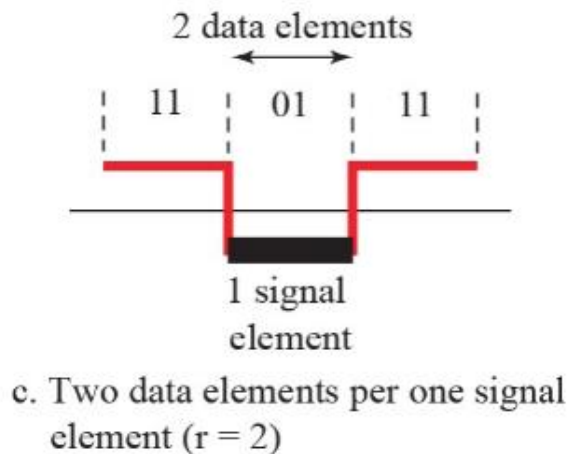
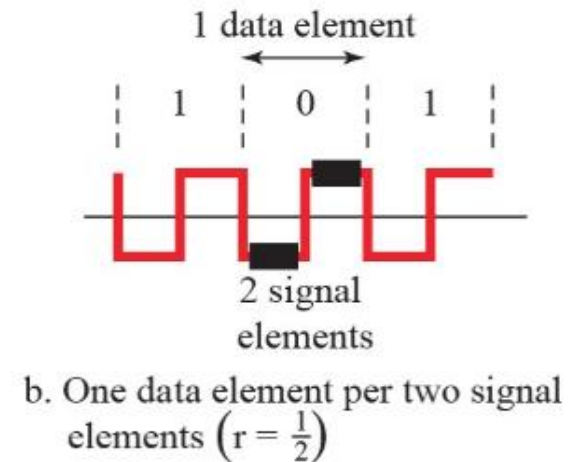
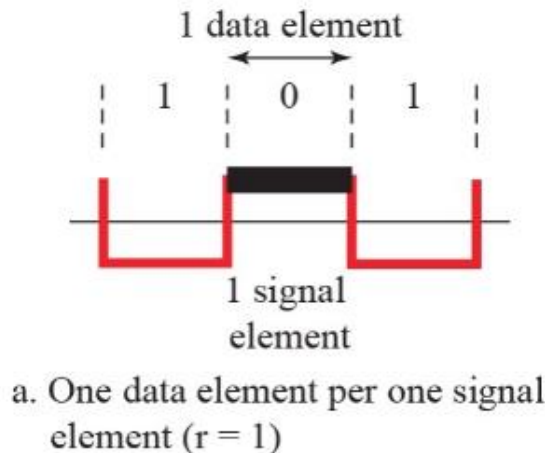
A ratio r is the number of data elements carried by each signal element.

$r = \text{No. of Data Elements} / \text{No. of Signal Elements}$

Line Coding

$r = \text{No. of Data Elements} / \text{No. of Signal Elements}$

Figure 4.2: Signal elements versus data elements



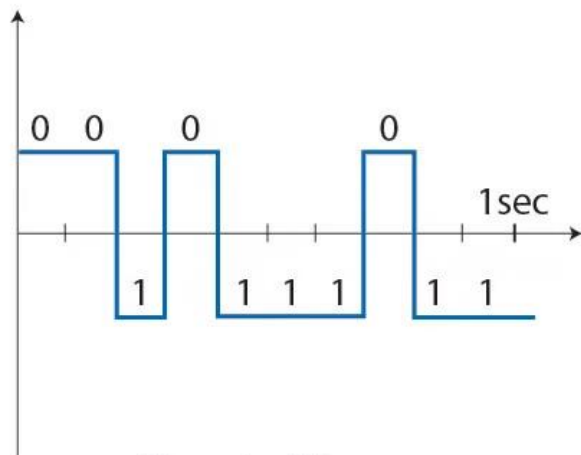
Line Coding

- In part a, one data element is carried by one signal element ($r = 1$).
- In part b, we need two signal elements (two transitions) to carry each data element ($r = 1/2$).
- In part c, a signal element carries two data elements ($r = 2$).
- In part d, a group of 4 bits is being carried by a group of three signal elements ($r = 4/3$).

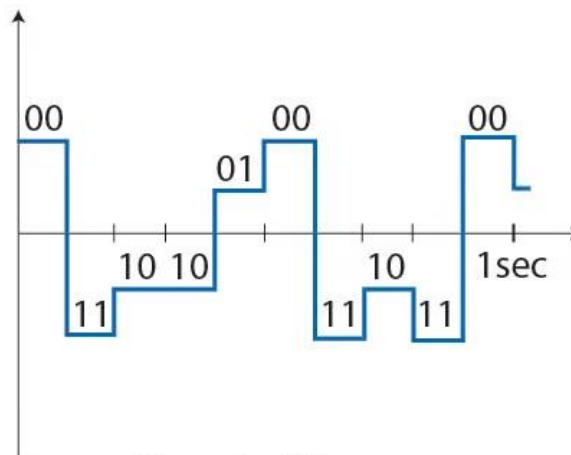
Line Coding

- ***Data Rate Versus Signal Rate***
- The **data rate** defines the number of data elements (bits) sent in 1s. The unit is bits per second (**bps**).
- The **signal rate** is the number of signal elements sent in 1s. The unit is the **baud**.
- The data rate is sometimes called the **bit rate**;
- the **signal rate** is sometimes called the
 - pulse rate,
 - modulation rate,
 - baud rate.

Line Coding



Baud = 10
Bit rate = 10 bps



Baud = 10
Bit rate = 20 bps

Line Coding

- *Goal* in data communications is to increase the data rate while decreasing the signal rate.
- **Increasing the data rate** increases the speed of transmission;
- **Decreasing the signal rate** decreases the bandwidth requirement.
- *Relationship between data rate (N) and signal rate (S)*

$$S = N/r$$

Line Coding

Three cases:

- **Worst case** is when we need the maximum signal rate.
- **Best case** is when we need the minimum signal rate.
- **Average case** is required in data communication
- We can formulate the **relationship between data rate and signal rate as**

$$S_{ave} = c \times N \times (1/r) \quad \text{baud}$$

Where,

- **N** is the data rate (bps);
- **c** is the case factor, which varies for each case: The value of the case factor (c) is not limited to specific numbers; it can take a wide range of values depending on the **design and operational requirements of the communication system**. The specific conditions and constraints of the system will determine the actual value of the case factor. It would take a value greater than 1 in a worst-case scenario, close to 1 in the ideal case, and less than in a practical scenario.
- **S** is the number of signal elements per second;
- **r** is the previously defined factor.

Line Coding

Example 4.1

A signal is carrying data in which one data element is encoded as one signal element ($r = 1$). If the bit rate is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?

Solution

We assume that the average value of c is $1/2$. The baud rate is then

$$S = c \times N \times (1 / r) = 1/2 \times 100,000 \times (1/1) = 50,000 = 50 \text{ kbaud}$$

□ □ □ □

Line Coding

- Baud rate determines the required bandwidth for a digital signal.
- Bandwidth is proportional to the signal rate (baud rate).
- The **minimum bandwidth** can be given as

$$B_{\min} = c \times N \times (1 / r)$$

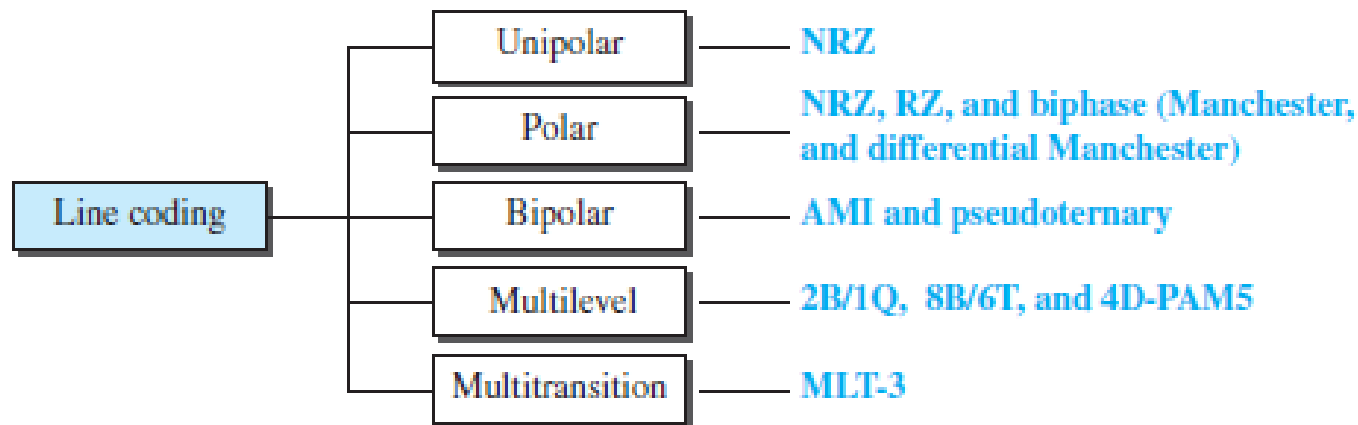
- **Maximum data rate** if the bandwidth of the channel is given.

$$N_{\max} = (1 / c) \times B \times r$$

Line Coding Schemes

We can roughly divide line coding schemes into five broad categories, as shown in Figure 4.4.

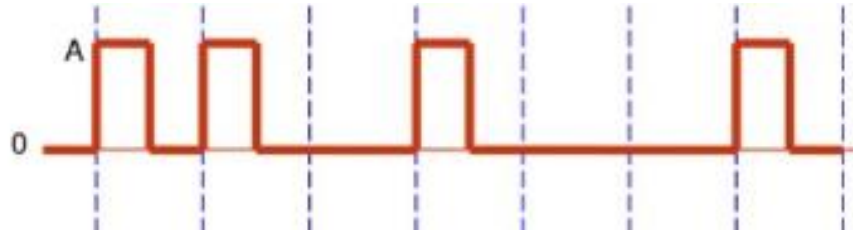
Figure 4.4 *Line coding schemes*



Line Coding Schemes

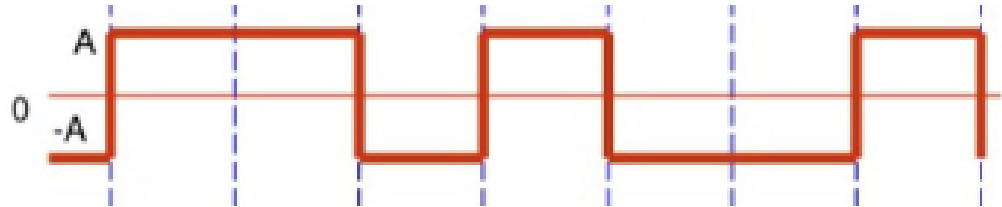
- **Unipolar scheme:** All signal levels are either below or above the axis.

Unipolar RZ



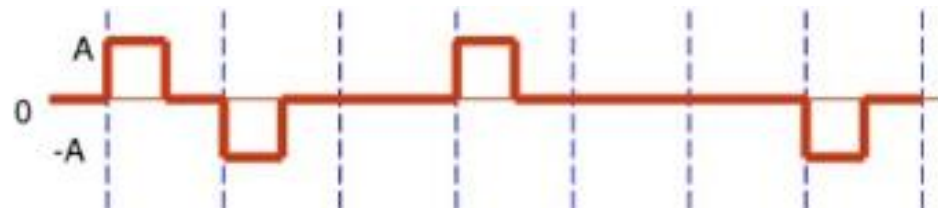
- **Polar Schemes:** Voltages are present on both sides of an axis.

Polar NRZ



- **Bipolar scheme:** There are three voltages: negative, positive, and zero.

Bipolar RZ



Line Coding Schemes

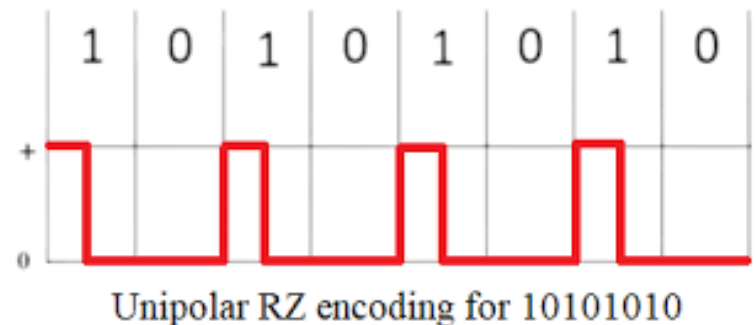
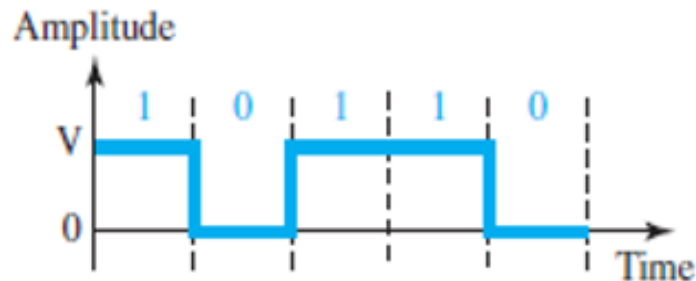
Unipolar Scheme

In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below.

NRZ (Non-Return-to-Zero)

Traditionally, a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit.

Figure 4.5 *Unipolar NRZ scheme*



Line Coding Schemes

Polar Schemes

In polar schemes, the voltages are on both sides of the time axis.

For example, the voltage level for 0 can be positive and the voltage level for 1 can be negative.

Non-Return-to-Zero (NRZ)

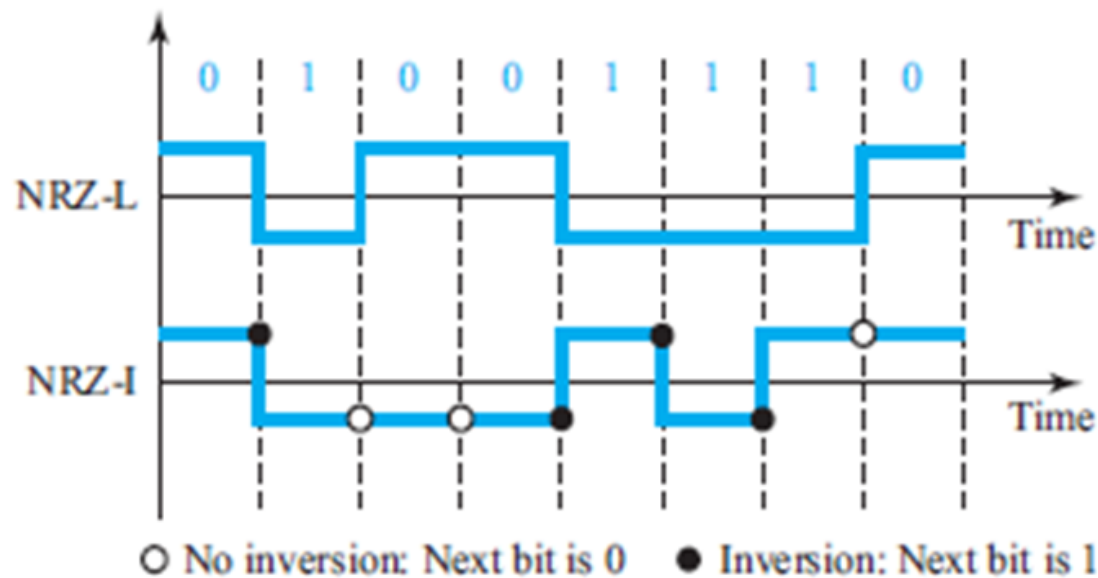
In polar NRZ encoding, we use two levels of voltage amplitude.

Two versions of polar NRZ: NRZ-L and NRZ-I.

Line Coding Schemes

- **NRZ-L (NRZ-Level)**, the level of the voltage determines the value of the bit.
- **NRZ-I (NRZ-Invert)**, the change or lack of change in the level of the voltage determines the value of the bit.
 - If there is no change, the bit is 0; if there is a change, the bit is 1.

Figure 4.6 *Polar NRZ-L and NRZ-I schemes*



Line Coding Schemes

Baseline Wandering

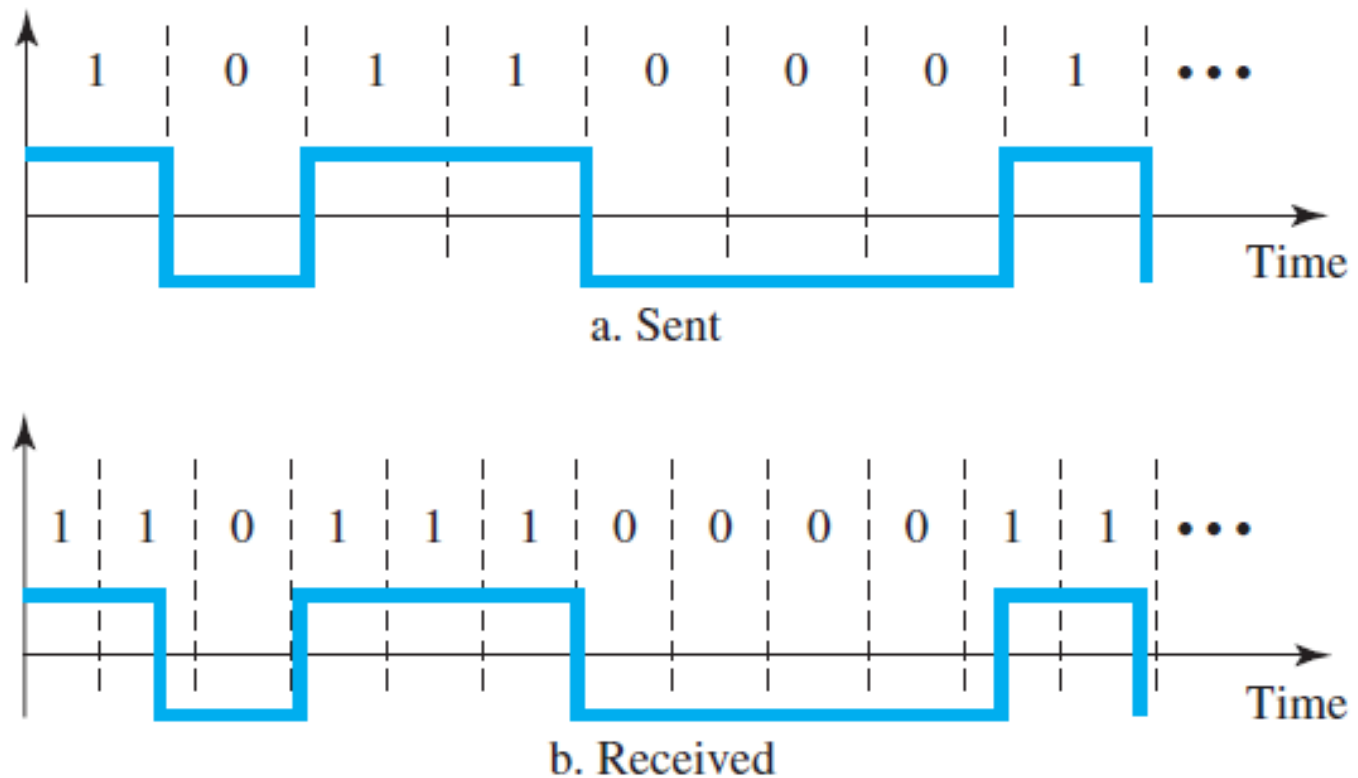
- In decoding a digital signal, the receiver calculates a running average of the received signal power. This average is called the ***baseline***.
- The incoming signal power is evaluated against this baseline to determine the value of the data element.
- A long string of 0s or 1s can cause a drift in the baseline (**baseline wandering**) and make it difficult for the receiver to decode correctly.
- A good line coding scheme needs to prevent baseline wandering.

Line Coding Schemes

- ***Self-synchronization***
- To correctly interpret the signals received from the sender, the receiver's bit intervals must correspond exactly to the sender's bit intervals.
- If the receiver clock is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals.
- If the receiver has a shorter bit duration. The sender sends 10110001, while the receiver receives 110111000011.
- A **self-synchronizing** digital signal includes timing information in the data being transmitted.
- This can be achieved if there are transitions in the signal that alert the receiver to the beginning, middle, or end of the pulse.
- If the receiver's clock is out of synchronization, these points can reset the clock.

Line Coding Schemes

Figure 4.3 *Effect of lack of synchronization*



Line Coding Schemes

Example 4.3

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 kbps? How many if the data rate is 1 Mbps?

Solution

At 1 kbps, the receiver receives 1001 bps instead of 1000 bps.

1000 bits sent → 1001 bits received → 1 extra bps

At 1 Mbps, the receiver receives 1,001,000 bps instead of 1,000,000 bps.

1,000,000 bits sent → 1,001,000 bits received → 1000 extra bps

Line Coding Schemes

Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level. Compare your guess with the corresponding entry in Table 4.1.

a. 00000000

b. 11111111

c. 01010101

d. 00110011

Line Coding Schemes

Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level. Compare your guess with the corresponding entry in Table 4.1.

a. 00000000

b. 11111111

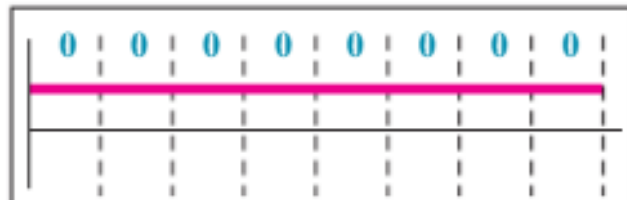
c. 01010101

d. 00110011

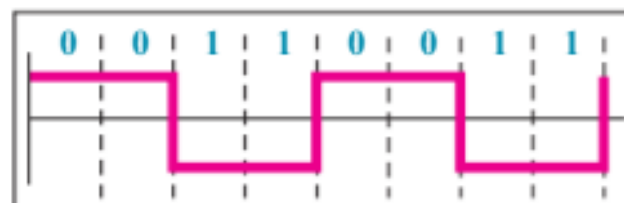
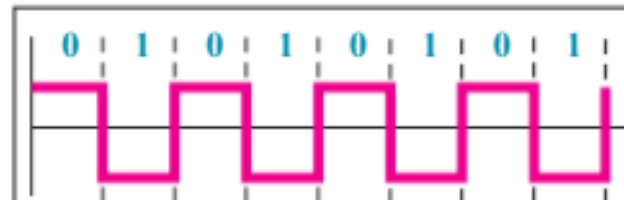
Average Number of Changes = $(0 + 0 + 8 + 4) / 4 = 3$ for $N = 8$

B $\rightarrow$ $(3 / 8) N$

Case a



Case c



Case b

Case d

Line Coding Schemes

Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level. Compare your guess with the corresponding entry in Table 4.1.

a. 00000000

b. 11111111

c. 01010101

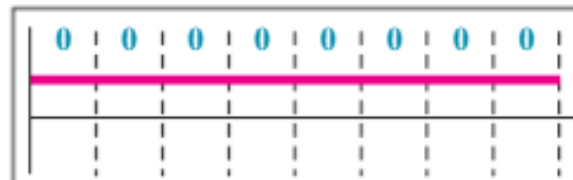
d. 00110011

NRZ-I

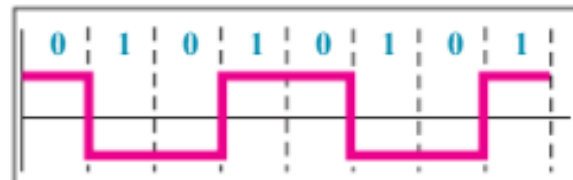
Average Number of Changes = $(0 + 9 + 4 + 4) / 4 = 4.25$ for $N = 8$

B $\rightarrow$ $(4.25 / 8) N$

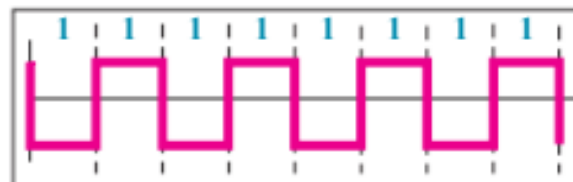
Case a



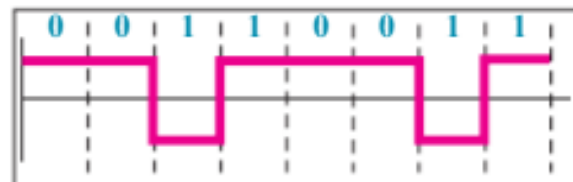
Case c



Case b



Case d

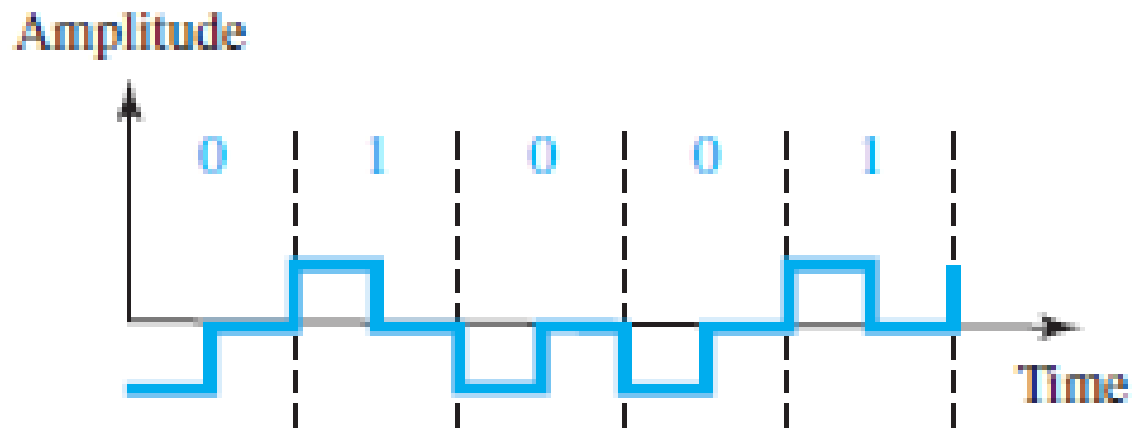


Line Coding Schemes: Polar-RZ

- The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized.
- The receiver does not know when one bit has ended and the next bit is starting.
- One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero.
- In RZ, the signal changes not between bits but during the bit.
- Signal goes to 0 in the middle of each bit. It remains there until the beginning of the next bit.

Line Coding Schemes: Polar-RZ

Figure 4.7 *Polar RZ scheme*



Line Coding Schemes: Polar-RZ

Disadvantages of RZ encoding

- Requires two signal changes to encode a bit and therefore occupies greater bandwidth.
- A sudden change of polarity resulting in all 0s interpreted as 1s and all 1s interpreted as 0s.
- RZ uses three levels of voltage, which is more complex to create and discern.
- As a result of all these deficiencies, the scheme is not used today. Instead, it has been replaced by the better-performing Manchester and differential Manchester schemes

Line Coding Schemes: Polar-RZ

Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level. Compare your guess with the corresponding entry in Table 4.1.

a. 00000000

b. 11111111

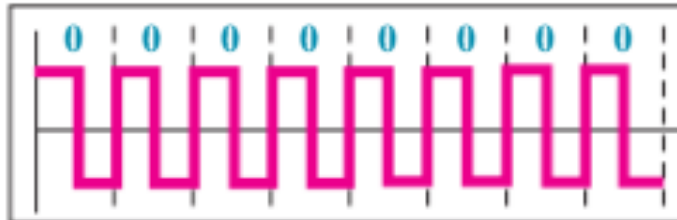
c. 01010101

d. 00110011

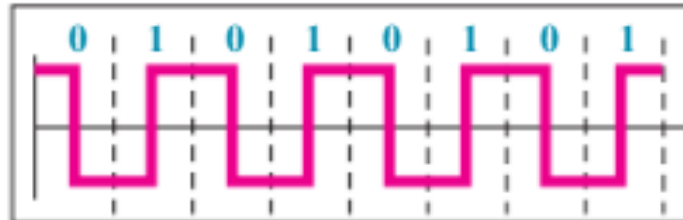
$$\text{Average Number of Changes} = (15 + 15 + 8 + 12) / 4 = 12.5 \text{ for } N = 8$$

B $\rightarrow$ $(12.5 / 8) N$

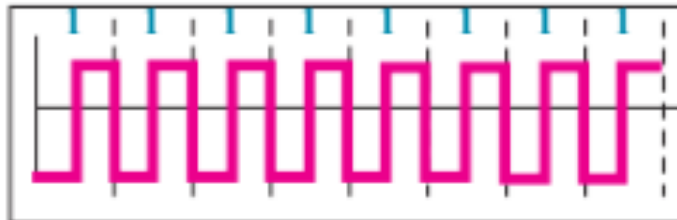
Case a



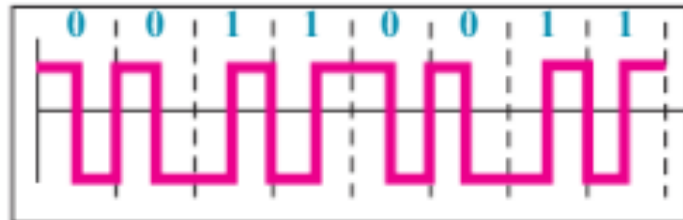
Case c



Case b



Case d



Line Coding Schemes: Polar : Biphas

Biphase: Manchester and Differential Manchester

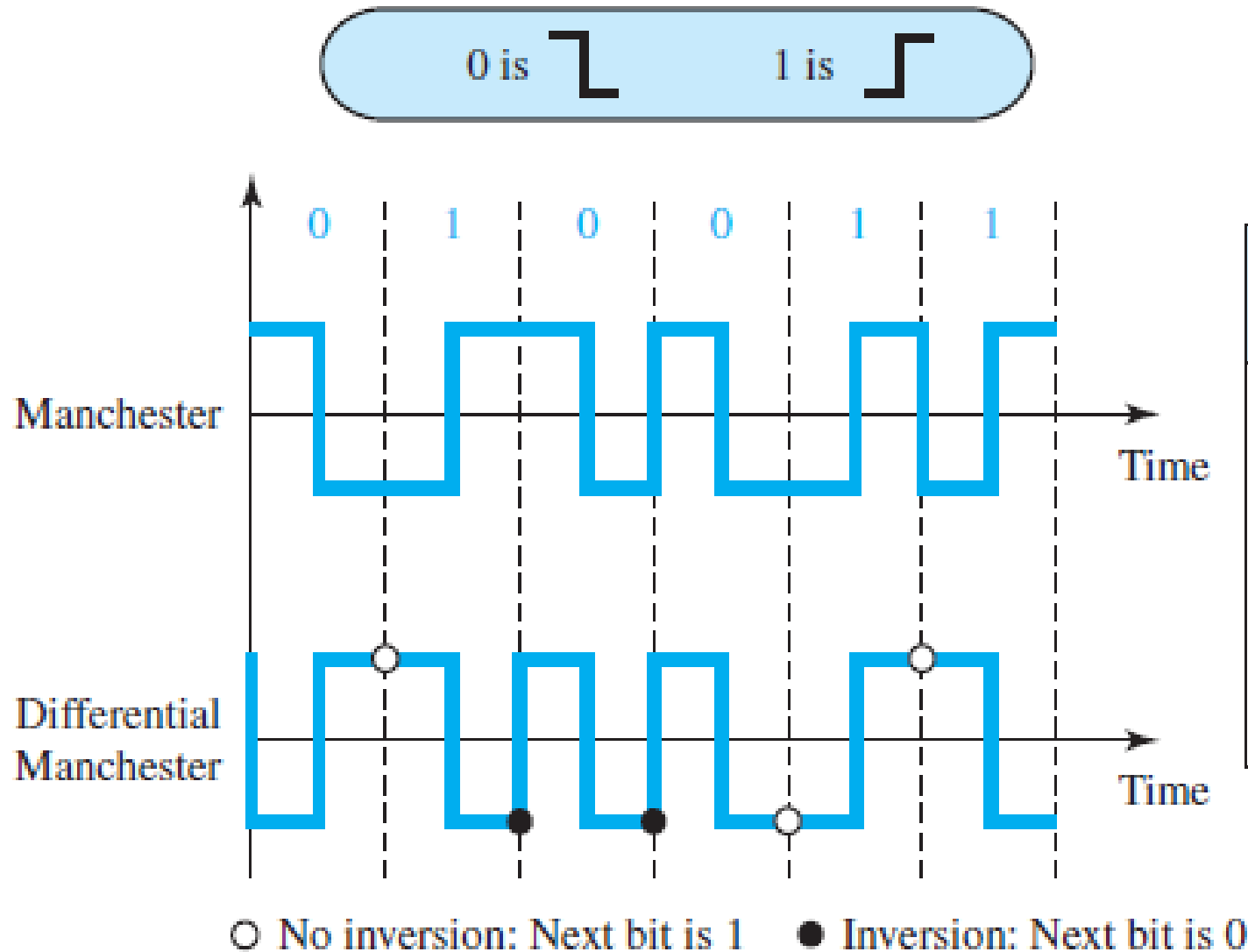
- The idea of RZ (transition at the middle of the bit) and the idea of NRZ-L are combined into the Manchester scheme.
- In Manchester encoding, the duration of the bit is divided into two halves.
- The voltage remains at one level during the first half and moves to the other level in the second half.
- The transition at the middle of the bit provides synchronization.

Line Coding Schemes: Polar-RZ

Differential Manchester

- Combines the ideas of RZ and NRZ-I.
- There is a transition at the middle of the bit, but the bit values are determined at the beginning of the bit.
- If the next bit is 0, there is a transition; if the next bit is 1, there is none.

Line Coding Schemes: Polar-RZ

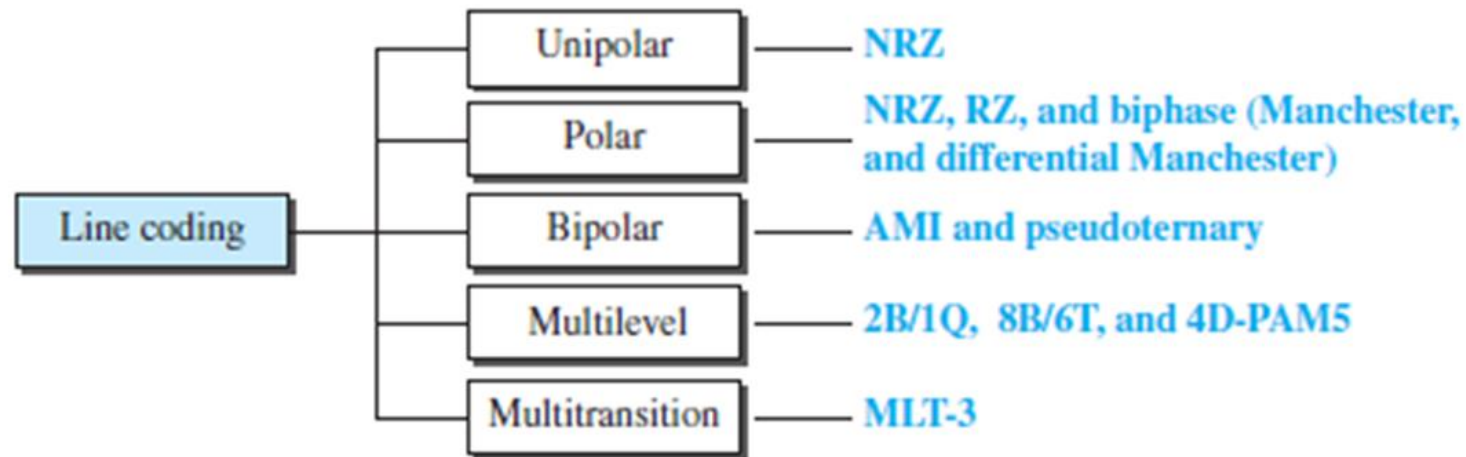


Line Coding Schemes: Polar-RZ

- The Manchester scheme overcomes several problems associated with NRZ-L
- Differential Manchester overcomes several problems associated with NRZ-I.
- **Drawback is the signal rate.** The signal rate for Manchester and differential Manchester is double that for NRZ.
- The reason is that there is always one transition at the middle of the bit and maybe one transition at the end of each bit.
- Manchester and differential Manchester schemes are also called **biphase schemes**.
- **The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.**

Line Coding Schemes: Polar-RZ

Figure 4.4 *Line coding schemes*



Line Coding Schemes: Bipolar

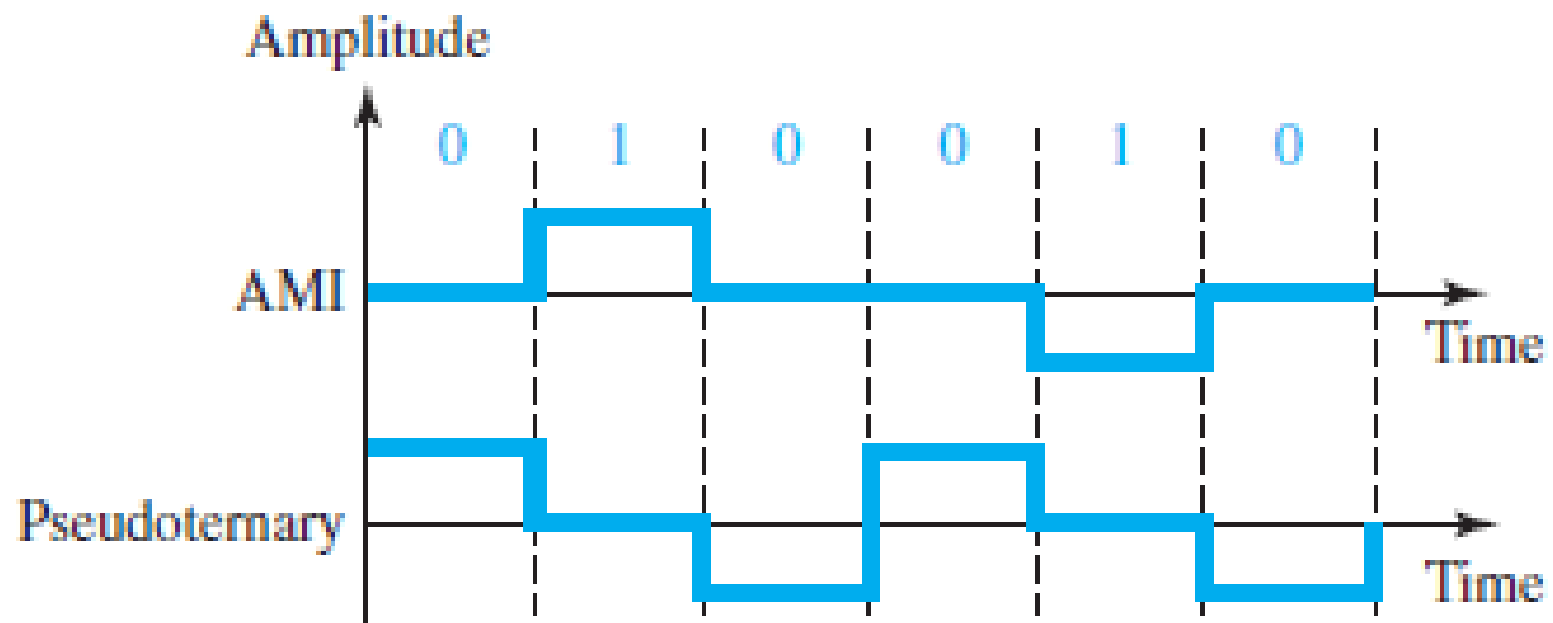
- In bipolar encoding, we use three levels: positive, zero, and negative.

AMI and Pseudoternary

- A common bipolar encoding scheme is called bipolar alternate mark inversion (AMI).
- In the term alternate mark inversion, the word mark comes from telegraphy and means 1.
- So AMI means alternate 1 inversion. A neutral zero voltage represents binary 0.
- Binary 1s are represented by alternating positive and negative voltages.
- A variation of AMI encoding is called pseudoternary in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.

Line Coding Schemes: Bipolar

Figure 4.9 *Bipolar schemes: AMI and pseudoternary*



Line Coding Schemes: Multitransition:MLT-3

Table 4.1 *Summary of line coding schemes*

<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC

ANALOG-TO-DIGITAL CONVERSION

- An analog signal such as one created by a microphone or camera.
- Digital signal is superior to an analog signal.
- Change an analog signal to digital data.
- Two techniques:
 1. Pulse Code Modulation (PCM)
 2. Delta Modulation.
- After the digital data are created (digitization), use one of the techniques described to convert the digital data to a digital signal

Pulse Code Modulation (PCM)

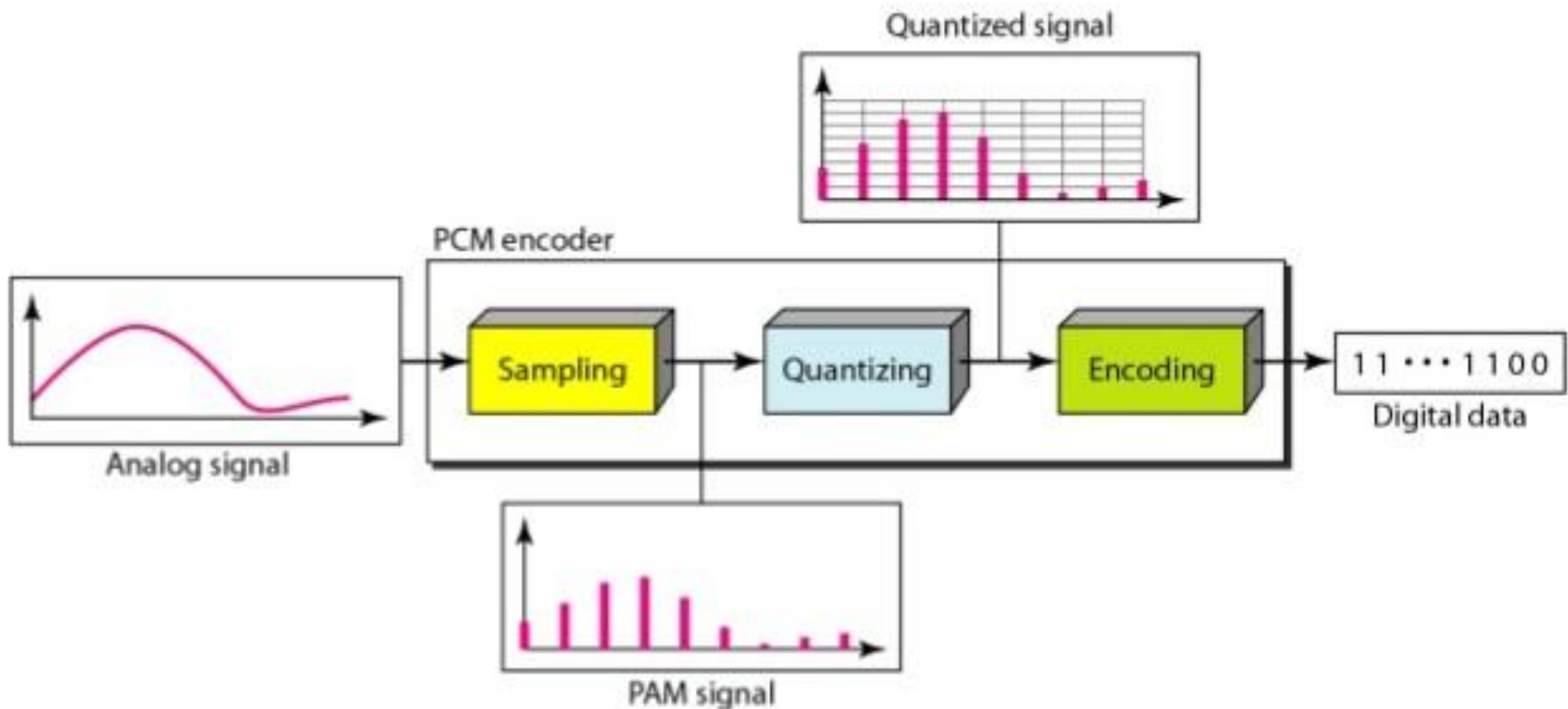
- An analog signal such as one created by a microphone or camera.
- Digital signal is superior to an analog signal.
- Change an analog signal to digital data.
- Two techniques:
 1. Pulse Code Modulation (PCM)
 2. Delta Modulation.
- After the digital data are created (digitization), use one of the techniques described to convert the digital data to a digital signal

Pulse Code Modulation (PCM)

The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).

A PCM encoder has three processes

1. The analog signal is **sampled**.
2. The sampled signal is **quantized**.
3. The quantized values are **encoded** as streams of bits.



Pulse Code Modulation (PCM)

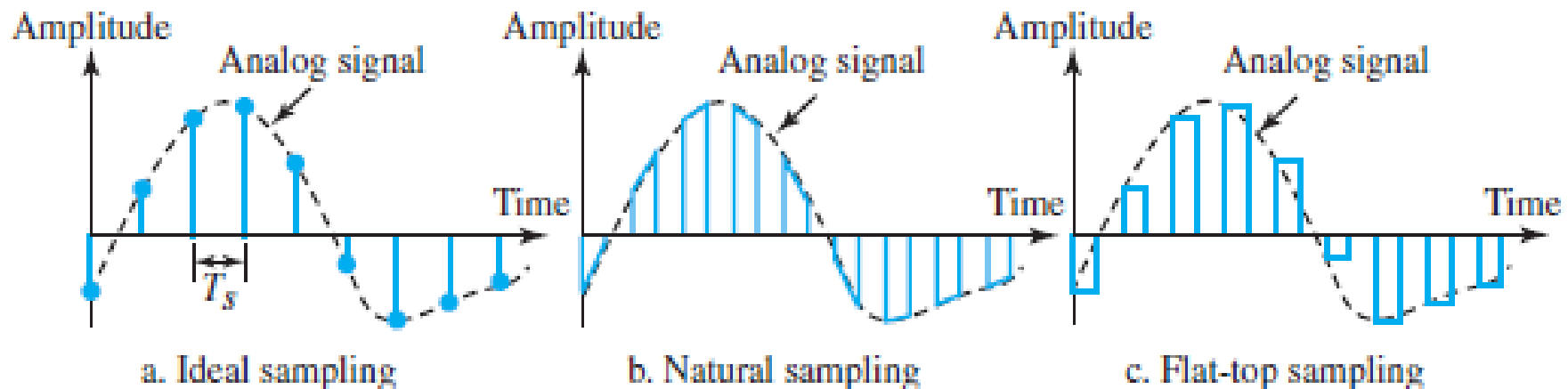
Sampling

- The analog signal is sampled every T_s s, where T_s is the sample interval or period.
- The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by f_s , where $f_s = 1/T_s$.
- There are three sampling methods—
ideal, natural, and flat-top.
- The sampling process is sometimes referred to as **pulse amplitude modulation (PAM).**

Pulse Code Modulation (PCM)

- In **ideal sampling**, pulses from the analog signal are sampled. This is an ideal sampling method and cannot be easily implemented.
- In **natural sampling**, a high-speed switch is turned on for only the small period of time when the sampling occurs. The result is a sequence of samples that retains the shape of the analog signal.
- The most common sampling method, called sample and hold, however, creates **flat-top samples** by using a circuit.

Figure 4.22 *Three different sampling methods for PCM*



Pulse Code Modulation (PCM)

Quantization

- The result of sampling is a series of pulses with **amplitude values between the maximum and minimum amplitudes of the signal**.
- The set of amplitudes can be infinite with **nonintegral values** between the two limits.
- These values cannot be used in the encoding process. The following are the steps in quantization:
 1. Assume that the original analog signal has instantaneous amplitudes between $V_{\min}$ and $V_{\max}$.
 2. Divide the range into L zones, each of height Δ (delta).

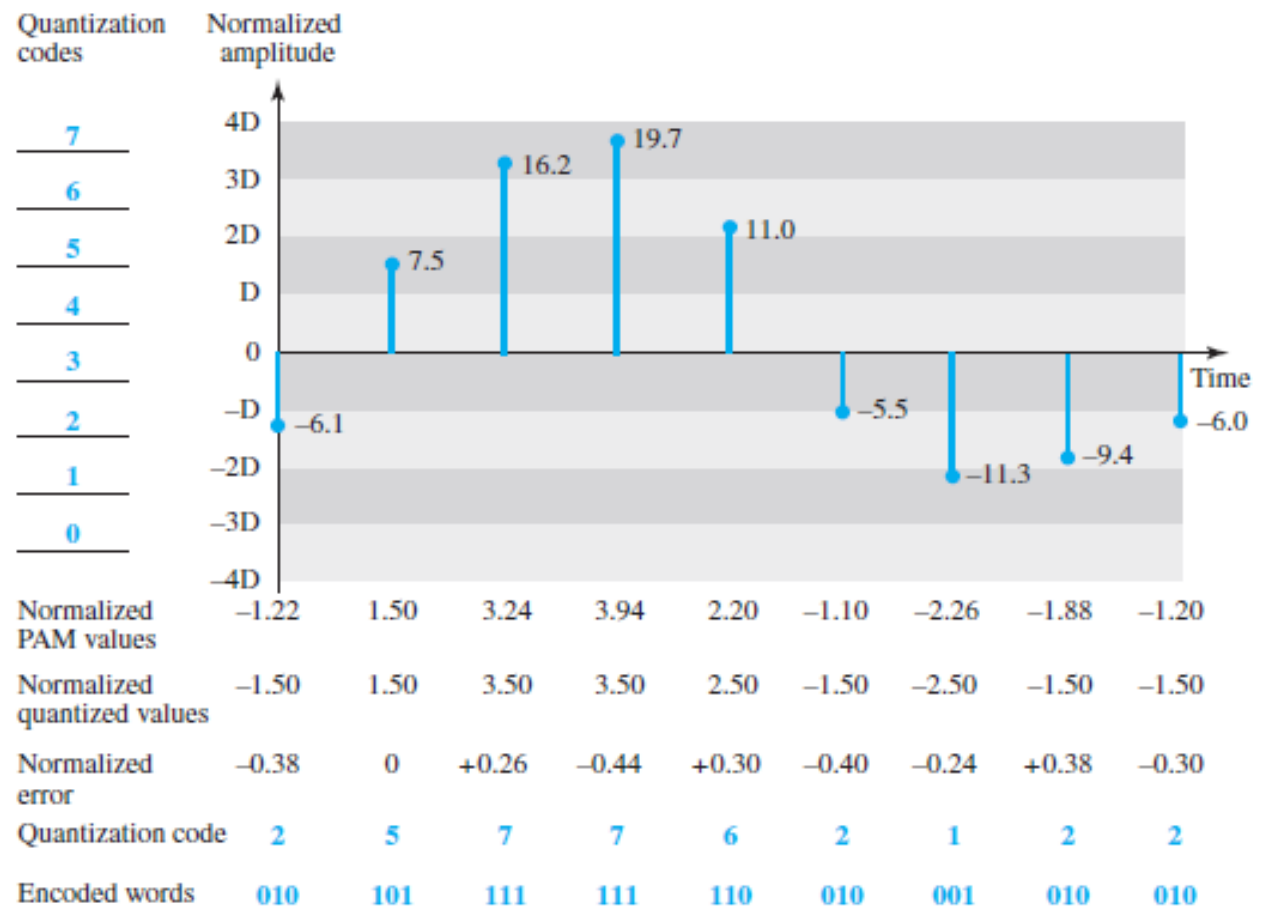
$$\Delta = \frac{V_{\max} - V_{\min}}{L}$$

3. Assign quantized values of 0 to $L - 1$ to the midpoint of each zone.
4. Approximate the value of the sample amplitude to the quantized values.

Pulse Code Modulation (PCM)

Example: Assume that we have a sampled signal and the sample amplitudes are between -20 and $+20$ V. We decide to have eight levels ($L = 8$). This means that $\Delta = 5$ V. Figure 4.26 shows this example

Figure 4.26 *Quantization and encoding of a sampled signal*



Pulse Code Modulation (PCM)

- The value at the top of each sample in the graph shows the **actual amplitude**.
- In the chart, the first row is the **normalized value** for each sample (**actual amplitude/ Δ**).
- The **quantization process** selects the quantization value from the middle of each zone.
- This means that the normalized quantized values (second row) are different from the normalized amplitudes.
- The difference is called the **normalized error (third row)**.
- The **fourth row** is the **quantization code** for each sample based on the quantization levels at the left of the graph.
- The **encoded words (fifth row)** are the final products of the conversion.

Pulse Code Modulation (PCM)

Quantization Levels

- The choice of L , the number of levels, depends on the range of the amplitudes of the analog signal and how accurately we need to recover the signal.
- If the amplitude of a signal fluctuates between two values only, we need only two levels.
- If the signal, like voice, has many amplitude values, we need more quantization levels.
- In audio digitizing, L is normally chosen to be 256; in video it is normally thousands.